Taehong Kwon
School of Computing, College of Science, Engineering and Technology
University of South Africa, South Africa
thkwon@enu-tech.co.kr

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The Editor-in-Chief
IEEE Access

Dear Editor,

We submit the manuscript "Coupling Authorization and Payment Edges in One PageRank Walk for Ranking Smart Contracts: Commit-Gated Rules and a Contract-Graph Re-Analysis on an Ethereum Rollup" for consideration as a Research Article in *IEEE Access*.

Wallets, block explorers and integrators need to know which of a rollup's more than a million smart contracts users rely on. Rankings of ledger addresses are usually computed from token payments, although the same addresses are also linked by token allowances, by which an owner authorizes a spender. The manuscript treats the two relations as layers of one network of contracts and tests, out of window, whether a single PageRank walk that mixes them ranks smart contracts on the Arbitrum One rollup better than a walk over either layer alone. Two decision rules were committed to the authors' Git repository before this study extracted the logs that judge them, the second eight seconds before its window was extracted, and the extraction code checks that the registration and configuration are committed without edits and records their SHA-256 hashes before it runs.

Neither rule was met, and the manuscript reports them as not met. Its main analysis was defined after the registered results were known: graphs whose nodes are contracts only, amounts in US dollars on five tokens, scores from the twelve months before each freeze, and each allowance counted up to its owner's balance. It is reported as a re-analysis under the same rules, beside the registered one. In both analyses the coupled walk ranked later approvals better than the payment walk but ranked later payers worse by more than the registered margin; on the contract graph it also ranked later approvals below the authorization walk. The paper states what the commit-gated record does and does not guarantee (commit identifiers, file hashes and cloud job times are given), and calibrates known link-farm results to an authorization layer whose nodes are contracts.

The work matters to an international readership. Graph-based ranking is often proposed for ordering the contracts of a chain; the paper shows on public data that on contract graphs the coupled walk beat neither of its layers, that plain counts of approving and of paying contracts did better than the coupled walk, that about 27% of the contracts to be ranked have no edge to another contract, and what a farm of contracts costs. *IEEE Access* reviews for technical soundness, which suits a study whose central findings are rules that were not met.

The manuscript is original, has not been published, and is not under consideration elsewhere. It is derived from the first author's doctoral research at the University of South Africa. A related manuscript by the same authors is being submitted at the same time to *PLOS ONE*. That paper reports a crawl-ordering test of whether walks reach the contracts users rely on sooner than counting, and sets the single-layer walks against the counts they smooth; the present paper concerns the coupling operator, its two pre-specified rules and their record, and the manipulability of the authorization layer. The two papers share the data and cite each other. Both report the spender holdout coefficients of the raw counts, EndorseRank and AWP with their count contrasts, the closed form of the authorization walk and the solver runtimes, attributed to the companion here. The crawl-ordering test appears only in the companion; the present paper quotes its verdict on the coupled walk. We upload the companion manuscript as confidential material for the reviewers. The study uses only public blockchain and price data and involves no human participants. The code, pre-specification documents, result summaries and the manuscript's scripts are public at github.com/abitocodes/ERC20-Allowance-PageRank-Wallet-Reputation and github.com/abitocodes/phd-dissertation-draft-en. The research received no external funding. The authors declare no competing interests; all have approved the submission.

Yours sincerely,

Taehong Kwon (corresponding author), on behalf of Ernest Mnkandla, Donatien Koulla Moulla and David Sena Attipoe